Modeling Neutron Stars
Lecture-6

Chiranjib Mondal
LPC, Université Caen Normandie
https://sites.google.com/view/nucleartheory/home

UNIVERSITÉ CAEN NORMANDIE

Nuclear Theory Group
January 30, 2021

Contents:

Elementary Field Theory

- Recap
- Infinite Nuclear matter
- Walecka's σ - ω Model
- Field Equations

Symmetries and Dirac Equation

Symmetries and Dirac Equation

- From "*stress-energy tensor*" we found out: $\boxed{T^{00} = \mathscr{H}}$

Symmetries and Dirac Equation

- From "*stress-energy tensor*" we found out: $\boxed{T^{00} = \mathscr{H}}$
- The Dirac equation is given for $\psi(x)$ as,

$$\boxed{(i\gamma^{\mu}\partial_{\mu} - m)\psi(x) = 0}$$

Symmetries and Dirac Equation

- From "*stress-energy tensor*" we found out: $\boxed{T^{00} = \mathscr{H}}$
- The Dirac equation is given for $\psi(x)$ as,

$$\boxed{(i\gamma^{\mu}\partial_{\mu} - m)\psi(x) = 0}$$

- The angular momentum operator intrinsically couples spin and orbit,

$$\boxed{J_{\mu\nu} = i(x_{\mu}\partial_{\nu} - x_{\nu}\partial_{\mu}) + \frac{1}{2}\sigma_{\mu\nu}}$$

Symmetries and Dirac Equation

- From "*stress-energy tensor*" we found out: $\boxed{T^{00} = \mathscr{H}}$
- The Dirac equation is given for $\psi(x)$ as,

$$\boxed{(i\gamma^\mu \partial_\mu - m)\psi(x) = 0}$$

- The angular momentum operator intrinsically couples spin and orbit,

$$\boxed{J_{\mu\nu} = i(x_\mu \partial_\nu - x_\nu \partial_\mu) + \frac{1}{2}\sigma_{\mu\nu}}$$

- A scalar under Lorentz transformation is given by $\boxed{\overline{\psi}\psi}$ and a vector is given by $\boxed{\overline{\psi}\gamma^\mu\psi}$.

Symmetries and Dirac Equation

- From "*stress-energy tensor*" we found out: $\boxed{T^{00} = \mathscr{H}}$
- The Dirac equation is given for $\psi(x)$ as,

$$\boxed{(i\gamma^\mu \partial_\mu - m)\psi(x) = 0}$$

- The angular momentum operator intrinsically couples spin and orbit,

$$\boxed{J_{\mu\nu} = i(x_\mu \partial_\nu - x_\nu \partial_\mu) + \frac{1}{2}\sigma_{\mu\nu}}$$

- A scalar under Lorentz transformation is given by $\boxed{\overline{\psi}\psi}$ and a vector is given by $\boxed{\overline{\psi}\gamma^\mu\psi}$.
- Lagrangian for free Dirac field is given by,

$$\boxed{\mathscr{L} = \overline{\psi}\left(i\gamma^\mu \partial_\mu - m\right)\psi}$$

Charge conservation

Dirac field

- $\mathscr{L}$ is invariant under the transformation

$$\boxed{\psi \longrightarrow e^{-iq\theta}\psi \Rightarrow \delta\psi = -iq\theta\psi}$$

Charge conservation

Dirac field

- $\mathscr{L}$ is invariant under the transformation

$$\boxed{\psi \longrightarrow e^{-iq\theta}\psi \Rightarrow \delta\psi = -iq\theta\psi}$$

- The corresponding Noether's current is given by

$$\boxed{j^{\mu} = i\overline{\psi}\gamma^{\mu}\frac{(-iq\theta\psi)}{\theta} = q\overline{\psi}\gamma^{\mu}\psi}.$$

Charge conservation

Dirac field

- $\mathscr{L}$ is invariant under the transformation

$$\boxed{\psi \longrightarrow e^{-iq\theta}\psi \Rightarrow \delta\psi = -iq\theta\psi}$$

- The corresponding Noether's current is given by

$$\boxed{j^{\mu} = i\overline{\psi}\gamma^{\mu}\frac{(-iq\theta\psi)}{\theta} = q\overline{\psi}\gamma^{\mu}\psi}.$$

$$\Rightarrow \boxed{Q = q\int d^3x\overline{\psi}\gamma^0\psi = q\int d^3x\psi^{\dagger}\psi}.$$

Nuclear Matter

Weizsacker's mass formula

- Idealized matter of infinite number of neutrons and protons without boundary and Coulomb interaction. Isotropic in nature.

Nuclear Matter

Weizsacker's mass formula

- Idealized matter of infinite number of neutrons and protons without boundary and Coulomb interaction. Isotropic in nature.
- Characterized by real nuclear data.

Nuclear Matter

Weizsacker's mass formula

- Idealized matter of infinite number of neutrons and protons without boundary and Coulomb interaction. Isotropic in nature.
- Characterized by real nuclear data.
- Saturated system

Nuclear Matter

Weizsacker's mass formula

- Idealized matter of infinite number of neutrons and protons without boundary and Coulomb interaction. Isotropic in nature.
- Characterized by real nuclear data.
- Saturated system $\longrightarrow$ short range and strong repulsion at short distance of the force.

Nuclear Matter

Weizsacker's mass formula

- Idealized matter of infinite number of neutrons and protons without boundary and Coulomb interaction. Isotropic in nature.
- Characterized by real nuclear data.
- Saturated system $\longrightarrow$ short range and strong repulsion at short distance of the force.
- From Weizsacker's mass formula $\rightarrow \frac{B}{A} \simeq -16$ MeV.

Nuclear Matter

Weizsacker's mass formula

- Idealized matter of infinite number of neutrons and protons without boundary and Coulomb interaction. Isotropic in nature.
- Characterized by real nuclear data.
- Saturated system $\longrightarrow$ short range and strong repulsion at short distance of the force.
- From Weizsacker's mass formula $\rightarrow \frac{B}{A} \simeq -16$ MeV.
- Radius of a nucleus can be parametrized by $R = r_0 A^{1/3}$.

Nuclear Matter

Weizsacker's mass formula

- Idealized matter of infinite number of neutrons and protons without boundary and Coulomb interaction. Isotropic in nature.
- Characterized by real nuclear data.
- Saturated system $\longrightarrow$ short range and strong repulsion at short distance of the force.
- From Weizsacker's mass formula $\rightarrow \frac{B}{A} \simeq -16$ MeV.
- Radius of a nucleus can be parametrized by $R = r_0 A^{1/3}$.
  $\Rightarrow r_0 \sim 1.15$ fm.

Nuclear Matter

Weizsacker's mass formula

- Idealized matter of infinite number of neutrons and protons without boundary and Coulomb interaction. Isotropic in nature.
- Characterized by real nuclear data.
- Saturated system $\longrightarrow$ short range and strong repulsion at short distance of the force.
- From Weizsacker's mass formula $\rightarrow \frac{B}{A} \simeq -16$ MeV.
- Radius of a nucleus can be parametrized by $R = r_0 A^{1/3}$.
  $\Rightarrow r_0 \sim 1.15$ fm.
- This can be translated into a saturation density

$$\rho_0 = \left[\left(\frac{4\pi}{3}\right) r_0^3\right]^{-1}$$

Nuclear Matter

Weizsacker's mass formula

- Idealized matter of infinite number of neutrons and protons without boundary and Coulomb interaction. Isotropic in nature.
- Characterized by real nuclear data.
- Saturated system $\longrightarrow$ short range and strong repulsion at short distance of the force.
- From Weizsacker's mass formula $\rightarrow \frac{B}{A} \simeq -16$ MeV.
- Radius of a nucleus can be parametrized by $R = r_0 A^{1/3}$.
  $\Rightarrow r_0 \sim 1.15$ fm.
- This can be translated into a saturation density

$$\rho_0 = \left[\left(\frac{4\pi}{3}\right) r_0^3\right]^{-1} \simeq 0.16 \text{ fm}^{-3}.$$

Nuclear Matter

Weizsacker's mass formula

- Idealized matter of infinite number of neutrons and protons without boundary and Coulomb interaction. Isotropic in nature.
- Characterized by real nuclear data.
- Saturated system $\longrightarrow$ short range and strong repulsion at short distance of the force.
- From Weizsacker's mass formula $\rightarrow \frac{B}{A} \simeq -16$ MeV.
- Radius of a nucleus can be parametrized by $R = r_0 A^{1/3}$.
  $\Rightarrow r_0 \sim 1.15$ fm.
- This can be translated into a saturation density

$$\rho_0 = \left[\left(\frac{4\pi}{3}\right) r_0^3\right]^{-1} \simeq 0.16 \text{ fm}^{-3}.$$

- Today our focus would be symmetric matter.

σ - ω Model

The Lagrangian

- Based on the fields of neutrons and protons (ψ field), scalar σ meson (like ϕ field) and vector ω^μ (like A^μ field with mass) meson.

σ - ω Model

The Lagrangian

- Based on the fields of neutrons and protons (ψ field), scalar σ meson (like ϕ field) and vector ω^μ (like A^μ field with mass) meson.
- The Lagrangian needs to be *Lorentz scalar*.

σ - ω Model

The Lagrangian

- Based on the fields of neutrons and protons (ψ field), scalar σ meson (like ϕ field) and vector ω^μ (like A^μ field with mass) meson.
- The Lagrangian needs to be *Lorentz scalar*. $\Rightarrow$ We couple σ field to $\overline{\psi}\psi$ and ω^μ to $\overline{\psi}\gamma^\mu\psi$.

σ - ω Model

The Lagrangian

- Based on the fields of neutrons and protons (ψ field), scalar σ meson (like ϕ field) and vector ω^μ (like A^μ field with mass) meson.
- The Lagrangian needs to be *Lorentz scalar*. $\Rightarrow$ We couple σ field to $\overline{\psi}\psi$ and ω^μ to $\overline{\psi}\gamma^\mu\psi$.
- The different Lagrangian terms can be written now:

σ - ω Model

The Lagrangian

- Based on the fields of neutrons and protons (ψ field), scalar σ meson (like ϕ field) and vector ω^μ (like A^μ field with mass) meson.
- The Lagrangian needs to be *Lorentz scalar*. $\Rightarrow$ We couple σ field to $\overline{\psi}\psi$ and ω^μ to $\overline{\psi}\gamma^\mu\psi$.
- The different Lagrangian terms can be written now:

$$\mathscr{L}_\sigma = \frac{1}{2}\left[(\partial_\mu\sigma)(\partial^\mu\sigma) - m_\sigma^2\sigma^2\right],$$

σ - ω Model

The Lagrangian

- Based on the fields of neutrons and protons (ψ field), scalar σ meson (like ϕ field) and vector ω^μ (like A^μ field with mass) meson.
- The Lagrangian needs to be *Lorentz scalar*. $\Rightarrow$ We couple σ field to $\overline{\psi}\psi$ and ω^μ to $\overline{\psi}\gamma^\mu\psi$.
- The different Lagrangian terms can be written now:

$$\begin{aligned}
\mathscr{L}_\sigma &= \frac{1}{2}\left[(\partial_\mu\sigma)(\partial^\mu\sigma) - m_\sigma^2\sigma^2\right], \\
\mathscr{L}_\omega &= -\frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2\omega_\mu\omega^\mu,
\end{aligned}$$

σ - ω Model

The Lagrangian

- Based on the fields of neutrons and protons (ψ field), scalar σ meson (like ϕ field) and vector ω^μ (like A^μ field with mass) meson.
- The Lagrangian needs to be *Lorentz scalar*. $\Rightarrow$ We couple σ field to $\overline{\psi}\psi$ and ω^μ to $\overline{\psi}\gamma^\mu\psi$.
- The different Lagrangian terms can be written now:

$$\begin{aligned}
\mathscr{L}_\sigma &= \frac{1}{2}\left[(\partial_\mu\sigma)(\partial^\mu\sigma) - m_\sigma^2\sigma^2\right], \\
\mathscr{L}_\omega &= -\frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2\omega_\mu\omega^\mu, \text{ where, } \Omega_{\mu\nu} = \partial_\mu\omega_\nu - \partial_\nu\omega_\mu,
\end{aligned}$$

σ - ω Model

The Lagrangian

- Based on the fields of neutrons and protons (ψ field), scalar σ meson (like ϕ field) and vector ω^μ (like A^μ field with mass) meson.
- The Lagrangian needs to be *Lorentz scalar*. $\Rightarrow$ We couple σ field to $\overline{\psi}\psi$ and ω^μ to $\overline{\psi}\gamma^\mu\psi$.
- The different Lagrangian terms can be written now:

$$\begin{aligned}
\mathscr{L}_\sigma &= \frac{1}{2}\left[(\partial_\mu\sigma)(\partial^\mu\sigma) - m_\sigma^2\sigma^2\right], \\
\mathscr{L}_\omega &= -\frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2\omega_\mu\omega^\mu, \text{ where, } \Omega_{\mu\nu} = \partial_\mu\omega_\nu - \partial_\nu\omega_\mu, \\
\mathscr{L}_\psi &= \overline{\psi}(i\gamma^\mu\partial_\mu - m)\psi,
\end{aligned}$$

σ - ω Model

The Lagrangian

- Based on the fields of neutrons and protons (ψ field), scalar σ meson (like ϕ field) and vector ω^μ (like A^μ field with mass) meson.
- The Lagrangian needs to be *Lorentz scalar*. $\Rightarrow$ We couple σ field to $\overline{\psi}\psi$ and ω^μ to $\overline{\psi}\gamma^\mu\psi$.
- The different Lagrangian terms can be written now:

$$
\begin{aligned}
\mathscr{L}_\sigma &= \frac{1}{2}\left[(\partial_\mu\sigma)(\partial^\mu\sigma) - m_\sigma^2\sigma^2\right], \\
\mathscr{L}_\omega &= -\frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2\omega_\mu\omega^\mu, \text{ where, } \Omega_{\mu\nu} = \partial_\mu\omega_\nu - \partial_\nu\omega_\mu, \\
\mathscr{L}_\psi &= \overline{\psi}(i\gamma^\mu\partial_\mu - m)\psi, \\
\mathscr{L}_{int} &= \mathrm{g}_\sigma\overline{\psi}\psi\sigma - \mathrm{g}_\omega\overline{\psi}\gamma^\mu\psi\omega_\mu,
\end{aligned}
$$

σ - ω Model

The Lagrangian

- Based on the fields of neutrons and protons (ψ field), scalar σ meson (like ϕ field) and vector ω^μ (like A^μ field with mass) meson.
- The Lagrangian needs to be *Lorentz scalar*. $\Rightarrow$ We couple σ field to $\overline{\psi}\psi$ and ω^μ to $\overline{\psi}\gamma^\mu\psi$.
- The different Lagrangian terms can be written now:

$$
\begin{aligned}
\mathscr{L}_\sigma &= \frac{1}{2}\left[(\partial_\mu\sigma)(\partial^\mu\sigma) - m_\sigma^2\sigma^2\right], \\
\mathscr{L}_\omega &= -\frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2\omega_\mu\omega^\mu, \text{ where, } \Omega_{\mu\nu} = \partial_\mu\omega_\nu - \partial_\nu\omega_\mu, \\
\mathscr{L}_\psi &= \overline{\psi}(i\gamma^\mu\partial_\mu - m)\psi, \\
\mathscr{L}_{int} &= \mathrm{g}_\sigma\overline{\psi}\psi\sigma - \mathrm{g}_\omega\overline{\psi}\gamma^\mu\psi\omega_\mu, \\
\mathscr{L} &= \mathscr{L}_\sigma + \mathscr{L}_\omega + \mathscr{L}_\psi + \mathscr{L}_{int}
\end{aligned}
$$

σ - ω Model

The Lagrangian

- Based on the fields of neutrons and protons (ψ field), scalar σ meson (like ϕ field) and vector ω^μ (like A^μ field with mass) meson.
- The Lagrangian needs to be *Lorentz scalar*. $\Rightarrow$ We couple σ field to $\overline{\psi}\psi$ and ω^μ to $\overline{\psi}\gamma^\mu\psi$.
- The different Lagrangian terms can be written now:

$$
\begin{aligned}
\mathscr{L}_\sigma &= \frac{1}{2}\left[(\partial_\mu\sigma)(\partial^\mu\sigma) - m_\sigma^2\sigma^2\right], \\
\mathscr{L}_\omega &= -\frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2\omega_\mu\omega^\mu, \text{ where, } \Omega_{\mu\nu} = \partial_\mu\omega_\nu - \partial_\nu\omega_\mu, \\
\mathscr{L}_\psi &= \overline{\psi}(i\gamma^\mu\partial_\mu - m)\psi, \\
\mathscr{L}_{int} &= \mathrm{g}_\sigma\overline{\psi}\psi\sigma - \mathrm{g}_\omega\overline{\psi}\gamma^\mu\psi\omega_\mu, \\
\mathscr{L} &= \mathscr{L}_\sigma + \mathscr{L}_\omega + \mathscr{L}_\psi + \mathscr{L}_{int} \\
&= \overline{\psi}\left[i\gamma^\mu\left(\partial_\mu + i\mathrm{g}_\omega\omega_\mu\right) - \left(m - \mathrm{g}_\sigma\sigma\right)\right]\psi \\
&\quad + \frac{1}{2}\left(\partial_\mu\sigma\partial^\mu\sigma - m_\sigma^2\sigma^2\right) - \frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2\omega_\mu\omega^\mu.
\end{aligned}
$$

Euler-Lagrange Equations

σ field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \sigma)} \right) = \frac{\partial \mathscr{L}}{\partial \sigma}$

Euler-Lagrange Equations

σ field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \sigma)} \right) = \frac{\partial \mathscr{L}}{\partial \sigma}$

- The free Lagrangian is given by

$$\mathscr{L}_\sigma = \frac{1}{2} \left[(\partial_\mu \sigma)(\partial^\mu \sigma) - m_\sigma^2 \sigma^2 \right]$$

Euler-Lagrange Equations

σ field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \sigma)} \right) = \frac{\partial \mathscr{L}}{\partial \sigma}$

- The free Lagrangian is given by

$$\mathscr{L}_\sigma = \frac{1}{2}\left[(\partial_\mu \sigma)(\partial^\mu \sigma) - m_\sigma^2 \sigma^2\right] = \frac{1}{2}\left[(\partial_\mu \sigma)(\partial_\nu \sigma) g^{\mu\nu} - m_\sigma^2 \sigma^2\right].$$

Euler-Lagrange Equations

σ field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \sigma)} \right) = \frac{\partial \mathscr{L}}{\partial \sigma}$

- The free Lagrangian is given by

$$\mathscr{L}_\sigma = \frac{1}{2}\left[(\partial_\mu \sigma)(\partial^\mu \sigma) - m_\sigma^2 \sigma^2\right] = \frac{1}{2}\left[(\partial_\mu \sigma)(\partial_\nu \sigma) g^{\mu\nu} - m_\sigma^2 \sigma^2\right].$$

- The corresponding Euler-Lagrange equation

$$\frac{\partial \mathscr{L}_\sigma}{\partial \sigma} = -m_\sigma^2 \sigma$$

Euler-Lagrange Equations

σ field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \sigma)} \right) = \frac{\partial \mathscr{L}}{\partial \sigma}$

- The free Lagrangian is given by

$$\mathscr{L}_\sigma = \frac{1}{2}\left[(\partial_\mu \sigma)(\partial^\mu \sigma) - m_\sigma^2 \sigma^2\right] = \frac{1}{2}\left[(\partial_\mu \sigma)(\partial_\nu \sigma) g^{\mu\nu} - m_\sigma^2 \sigma^2\right].$$

- The corresponding Euler-Lagrange equation

$$\frac{\partial \mathscr{L}_\sigma}{\partial \sigma} = -m_\sigma^2 \sigma$$

$$\partial_\alpha \left(\frac{\partial \mathcal{L}_\sigma}{\partial(\partial_\alpha \sigma)} \right) = \frac{1}{2}\left[\partial_\alpha (\delta_\mu^\alpha (\partial_\nu \sigma) g^{\mu\nu} + (\partial_\mu \sigma)\delta_\nu^\alpha g^{\mu\nu})\right]$$

Euler-Lagrange Equations

σ field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \sigma)} \right) = \frac{\partial \mathscr{L}}{\partial \sigma}$

- The free Lagrangian is given by

$$\mathscr{L}_\sigma = \frac{1}{2}\left[(\partial_\mu\sigma)(\partial^\mu\sigma) - m_\sigma^2\sigma^2\right] = \frac{1}{2}\left[(\partial_\mu\sigma)(\partial_\nu\sigma)g^{\mu\nu} - m_\sigma^2\sigma^2\right].$$

- The corresponding Euler-Lagrange equation

$$\begin{aligned}
\frac{\partial \mathscr{L}_\sigma}{\partial \sigma} &= -m_\sigma^2\sigma \\
\partial_\alpha\left(\frac{\partial \mathcal{L}_\sigma}{\partial(\partial_\alpha\sigma)}\right) &= \frac{1}{2}\left[\partial_\alpha(\delta^\alpha_\mu(\partial_\nu\sigma)g^{\mu\nu} + (\partial_\mu\sigma)\delta^\alpha_\nu g^{\mu\nu})\right] \\
&= \frac{1}{2}\left[\partial_\alpha\left\{(\partial^\alpha\sigma) + (\partial^\alpha\sigma)\right\}\right]
\end{aligned}$$

Euler-Lagrange Equations

σ field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \sigma)} \right) = \frac{\partial \mathscr{L}}{\partial \sigma}$

- The free Lagrangian is given by

$$\mathscr{L}_\sigma = \frac{1}{2}\left[(\partial_\mu \sigma)(\partial^\mu \sigma) - m_\sigma^2 \sigma^2\right] = \frac{1}{2}\left[(\partial_\mu \sigma)(\partial_\nu \sigma) g^{\mu\nu} - m_\sigma^2 \sigma^2\right].$$

- The corresponding Euler-Lagrange equation

$$\begin{aligned} \frac{\partial \mathscr{L}_\sigma}{\partial \sigma} &= -m_\sigma^2 \sigma \\ \partial_\alpha \left(\frac{\partial \mathcal{L}_\sigma}{\partial(\partial_\alpha \sigma)} \right) &= \frac{1}{2}\left[\partial_\alpha (\delta_\mu^\alpha (\partial_\nu \sigma) g^{\mu\nu} + (\partial_\mu \sigma)\delta_\nu^\alpha g^{\mu\nu})\right] \\ &= \frac{1}{2}\left[\partial_\alpha \left\{(\partial^\alpha \sigma) + (\partial^\alpha \sigma)\right\}\right] = \partial_\alpha \partial^\alpha \sigma \end{aligned}$$

Euler-Lagrange Equations

σ field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \sigma)} \right) = \frac{\partial \mathscr{L}}{\partial \sigma}$

- The free Lagrangian is given by

$$\mathscr{L}_\sigma = \frac{1}{2}\left[(\partial_\mu\sigma)(\partial^\mu\sigma) - m_\sigma^2\sigma^2\right] = \frac{1}{2}\left[(\partial_\mu\sigma)(\partial_\nu\sigma)g^{\mu\nu} - m_\sigma^2\sigma^2\right].$$

- The corresponding Euler-Lagrange equation

$$\begin{aligned}
\frac{\partial \mathscr{L}_\sigma}{\partial \sigma} &= -m_\sigma^2\sigma \\
\partial_\alpha\left(\frac{\partial \mathcal{L}_\sigma}{\partial(\partial_\alpha\sigma)}\right) &= \frac{1}{2}\left[\partial_\alpha(\delta^\alpha_\mu(\partial_\nu\sigma)g^{\mu\nu} + (\partial_\mu\sigma)\delta^\alpha_\nu g^{\mu\nu})\right] \\
&= \frac{1}{2}\left[\partial_\alpha\left\{(\partial^\alpha\sigma) + (\partial^\alpha\sigma)\right\}\right] = \partial_\alpha\partial^\alpha\sigma = \Box\sigma.
\end{aligned}$$

Euler-Lagrange Equations

σ field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \sigma)} \right) = \frac{\partial \mathscr{L}}{\partial \sigma}$

- The free Lagrangian is given by

$$\mathscr{L}_\sigma = \frac{1}{2}\left[(\partial_\mu\sigma)(\partial^\mu\sigma) - m_\sigma^2\sigma^2\right] = \frac{1}{2}\left[(\partial_\mu\sigma)(\partial_\nu\sigma)g^{\mu\nu} - m_\sigma^2\sigma^2\right].$$

- The corresponding Euler-Lagrange equation

$$\begin{aligned} \frac{\partial \mathscr{L}_\sigma}{\partial \sigma} &= -m_\sigma^2\sigma \\ \partial_\alpha\left(\frac{\partial \mathcal{L}_\sigma}{\partial(\partial_\alpha\sigma)}\right) &= \frac{1}{2}\left[\partial_\alpha(\delta^\alpha_\mu(\partial_\nu\sigma)g^{\mu\nu} + (\partial_\mu\sigma)\delta^\alpha_\nu g^{\mu\nu})\right] \\ &= \frac{1}{2}\left[\partial_\alpha\left\{(\partial^\alpha\sigma) + (\partial^\alpha\sigma)\right\}\right] = \partial_\alpha\partial^\alpha\sigma = \Box\sigma. \end{aligned}$$

- Contributions also comes from $\mathscr{L}_{int}$.

Euler-Lagrange Equations

σ field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \sigma)} \right) = \frac{\partial \mathscr{L}}{\partial \sigma}$

- The free Lagrangian is given by

$$\mathscr{L}_\sigma = \frac{1}{2}\left[(\partial_\mu\sigma)(\partial^\mu\sigma) - m_\sigma^2\sigma^2\right] = \frac{1}{2}\left[(\partial_\mu\sigma)(\partial_\nu\sigma)g^{\mu\nu} - m_\sigma^2\sigma^2\right].$$

- The corresponding Euler-Lagrange equation

$$\begin{aligned}
\frac{\partial \mathscr{L}_\sigma}{\partial \sigma} &= -m_\sigma^2\sigma \\
\partial_\alpha\left(\frac{\partial \mathcal{L}_\sigma}{\partial(\partial_\alpha\sigma)}\right) &= \frac{1}{2}\left[\partial_\alpha(\delta^\alpha_\mu(\partial_\nu\sigma)g^{\mu\nu} + (\partial_\mu\sigma)\delta^\alpha_\nu g^{\mu\nu})\right] \\
&= \frac{1}{2}\left[\partial_\alpha\left\{(\partial^\alpha\sigma) + (\partial^\alpha\sigma)\right\}\right] = \partial_\alpha\partial^\alpha\sigma = \Box\sigma.
\end{aligned}$$

- Contributions also comes from $\mathscr{L}_{int}$.

$$\frac{\partial \mathscr{L}_{int}}{\partial \sigma} = \frac{\partial\left(\mathrm{g}_\sigma\overline{\psi}\psi\sigma\right)}{\partial\sigma}$$

Euler-Lagrange Equations

σ field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \sigma)} \right) = \frac{\partial \mathscr{L}}{\partial \sigma}$

- The free Lagrangian is given by

$$\mathscr{L}_\sigma = \frac{1}{2}\left[(\partial_\mu\sigma)(\partial^\mu\sigma) - m_\sigma^2\sigma^2\right] = \frac{1}{2}\left[(\partial_\mu\sigma)(\partial_\nu\sigma)g^{\mu\nu} - m_\sigma^2\sigma^2\right].$$

- The corresponding Euler-Lagrange equation

$$\begin{aligned}
\frac{\partial \mathscr{L}_\sigma}{\partial \sigma} &= -m_\sigma^2\sigma \\
\partial_\alpha\left(\frac{\partial \mathcal{L}_\sigma}{\partial(\partial_\alpha\sigma)}\right) &= \frac{1}{2}\left[\partial_\alpha(\delta^\alpha_\mu(\partial_\nu\sigma)g^{\mu\nu} + (\partial_\mu\sigma)\delta^\alpha_\nu g^{\mu\nu})\right] \\
&= \frac{1}{2}\left[\partial_\alpha\left\{(\partial^\alpha\sigma) + (\partial^\alpha\sigma)\right\}\right] = \partial_\alpha\partial^\alpha\sigma = \Box\sigma.
\end{aligned}$$

- Contributions also comes from $\mathscr{L}_{int}$.

$$\frac{\partial \mathscr{L}_{int}}{\partial \sigma} = \frac{\partial\left(\mathrm{g}_\sigma\overline{\psi}\psi\sigma\right)}{\partial\sigma} = \mathrm{g}_\sigma\overline{\psi}\psi$$

Euler-Lagrange Equations

σ field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \sigma)} \right) = \frac{\partial \mathscr{L}}{\partial \sigma}$

- The free Lagrangian is given by

$$\mathscr{L}_\sigma = \frac{1}{2}\left[(\partial_\mu\sigma)(\partial^\mu\sigma) - m_\sigma^2\sigma^2\right] = \frac{1}{2}\left[(\partial_\mu\sigma)(\partial_\nu\sigma)g^{\mu\nu} - m_\sigma^2\sigma^2\right].$$

- The corresponding Euler-Lagrange equation

$$\begin{aligned}
\frac{\partial \mathscr{L}_\sigma}{\partial \sigma} &= -m_\sigma^2\sigma \\
\partial_\alpha\left(\frac{\partial \mathcal{L}_\sigma}{\partial(\partial_\alpha\sigma)}\right) &= \frac{1}{2}\left[\partial_\alpha(\delta^\alpha_\mu(\partial_\nu\sigma)g^{\mu\nu} + (\partial_\mu\sigma)\delta^\alpha_\nu g^{\mu\nu})\right] \\
&= \frac{1}{2}\left[\partial_\alpha\left\{(\partial^\alpha\sigma) + (\partial^\alpha\sigma)\right\}\right] = \partial_\alpha\partial^\alpha\sigma = \Box\sigma.
\end{aligned}$$

- Contributions also comes from $\mathscr{L}_{int}$.

$$\frac{\partial \mathscr{L}_{int}}{\partial \sigma} = \frac{\partial\left(\mathrm{g}_\sigma\overline{\psi}\psi\sigma\right)}{\partial\sigma} = \mathrm{g}_\sigma\overline{\psi}\psi \Rightarrow \boxed{(\Box + m_\sigma^2)\sigma = \mathrm{g}_\sigma\overline{\psi}\psi}.$$

Euler-Lagrange Equations

ω_ν field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial \left(\partial_\mu \omega_\nu \right)} \right) = \frac{\partial \mathscr{L}}{\partial \omega_\nu}$

Euler-Lagrange Equations

ω_ν field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \omega_\nu)} \right) = \frac{\partial \mathscr{L}}{\partial \omega_\nu}$

- The free Lagrangian is given by

$$\mathscr{L}_\omega = -\frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2 \omega_\mu \omega^\mu$$

Euler-Lagrange Equations

ω_ν field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \omega_\nu)} \right) = \frac{\partial \mathscr{L}}{\partial \omega_\nu}$

- The free Lagrangian is given by

$$\mathscr{L}_\omega = -\frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2 \omega_\mu \omega^\mu = -\frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2 \omega_\mu \omega_\alpha g^{\alpha\mu}$$

Euler-Lagrange Equations

ω_ν field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \omega_\nu)} \right) = \frac{\partial \mathscr{L}}{\partial \omega_\nu}$

- The free Lagrangian is given by

$$\mathscr{L}_\omega = -\frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2 \omega_\mu \omega^\mu = -\frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2 \omega_\mu \omega_\alpha g^{\alpha\mu}$$

- Following exactly the same technique of field strength tensor $F_{\mu\nu}$ of Lec-3, one can write

$$\partial_\mu \left(\frac{\partial \mathscr{L}_\omega}{\partial \left(\partial_\mu \omega_\nu\right)} \right) = -\partial_\mu \Omega^{\mu\nu}$$

Euler-Lagrange Equations

ω_ν field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \omega_\nu)} \right) = \frac{\partial \mathscr{L}}{\partial \omega_\nu}$

- The free Lagrangian is given by

$$\mathscr{L}_\omega = -\frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2 \omega_\mu \omega^\mu = -\frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2 \omega_\mu \omega_\alpha g^{\alpha\mu}$$

- Following exactly the same technique of field strength tensor $F_{\mu\nu}$ of Lec-3, one can write

$$\begin{aligned} \partial_\mu \left(\frac{\partial \mathscr{L}_\omega}{\partial(\partial_\mu \omega_\nu)} \right) &= -\partial_\mu \Omega^{\mu\nu} \\ \frac{\partial \mathscr{L}_\omega}{\partial \omega_\nu} &= \frac{1}{2}m_\omega^2 (\delta_\mu^\nu \omega_\alpha g^{\alpha\mu} + \delta_\alpha^\nu \omega_\mu g^{\alpha\mu}) \end{aligned}$$

Euler-Lagrange Equations

ω_ν field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \omega_\nu)} \right) = \frac{\partial \mathscr{L}}{\partial \omega_\nu}$

- The free Lagrangian is given by

$$\mathscr{L}_\omega = -\frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2 \omega_\mu \omega^\mu = -\frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2 \omega_\mu \omega_\alpha g^{\alpha\mu}$$

- Following exactly the same technique of field strength tensor $F_{\mu\nu}$ of Lec-3, one can write

$$\begin{aligned} \partial_\mu \left(\frac{\partial \mathscr{L}_\omega}{\partial(\partial_\mu \omega_\nu)} \right) &= -\partial_\mu \Omega^{\mu\nu} \\ \frac{\partial \mathscr{L}_\omega}{\partial \omega_\nu} &= \frac{1}{2}m_\omega^2 (\delta_\mu^\nu \omega_\alpha g^{\alpha\mu} + \delta_\alpha^\nu \omega_\mu g^{\alpha\mu}) = m_\omega^2 \omega^\nu \end{aligned}$$

Euler-Lagrange Equations

ω_ν field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \omega_\nu)} \right) = \frac{\partial \mathscr{L}}{\partial \omega_\nu}$

- The free Lagrangian is given by

$$\mathscr{L}_\omega = -\frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2 \omega_\mu \omega^\mu = -\frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2 \omega_\mu \omega_\alpha g^{\alpha\mu}$$

- Following exactly the same technique of field strength tensor $F_{\mu\nu}$ of Lec-3, one can write

$$\begin{aligned} \partial_\mu \left(\frac{\partial \mathscr{L}_\omega}{\partial(\partial_\mu \omega_\nu)} \right) &= -\partial_\mu \Omega^{\mu\nu} \\ \frac{\partial \mathscr{L}_\omega}{\partial \omega_\nu} &= \frac{1}{2}m_\omega^2 (\delta_\mu^\nu \omega_\alpha g^{\alpha\mu} + \delta_\alpha^\nu \omega_\mu g^{\alpha\mu}) = m_\omega^2 \omega^\nu \end{aligned}$$

- Contributions also comes from $\mathscr{L}_{int}$.

Euler-Lagrange Equations

ω_ν field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \omega_\nu)} \right) = \frac{\partial \mathscr{L}}{\partial \omega_\nu}$

- The free Lagrangian is given by

$$\mathscr{L}_\omega = -\frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2 \omega_\mu \omega^\mu = -\frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2 \omega_\mu \omega_\alpha g^{\alpha\mu}$$

- Following exactly the same technique of field strength tensor $F_{\mu\nu}$ of Lec-3, one can write

$$\begin{aligned} \partial_\mu \left(\frac{\partial \mathscr{L}_\omega}{\partial(\partial_\mu \omega_\nu)} \right) &= -\partial_\mu \Omega^{\mu\nu} \\ \frac{\partial \mathscr{L}_\omega}{\partial \omega_\nu} &= \frac{1}{2}m_\omega^2 (\delta_\mu^\nu \omega_\alpha g^{\alpha\mu} + \delta_\alpha^\nu \omega_\mu g^{\alpha\mu}) = m_\omega^2 \omega^\nu \end{aligned}$$

- Contributions also comes from $\mathscr{L}_{int}$.

$$\frac{\partial \mathscr{L}_{int}}{\partial \omega_\nu} = \frac{\partial \left(-\mathrm{g}_\omega \overline{\psi} \gamma^\mu \psi \omega_\mu \right)}{\partial \omega_\nu}$$

Euler-Lagrange Equations

ω_ν field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \omega_\nu)} \right) = \frac{\partial \mathscr{L}}{\partial \omega_\nu}$

- The free Lagrangian is given by

$$\mathscr{L}_\omega = -\frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2 \omega_\mu \omega^\mu = -\frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2 \omega_\mu \omega_\alpha g^{\alpha\mu}$$

- Following exactly the same technique of field strength tensor $F_{\mu\nu}$ of Lec-3, one can write

$$\begin{aligned} \partial_\mu \left(\frac{\partial \mathscr{L}_\omega}{\partial(\partial_\mu \omega_\nu)} \right) &= -\partial_\mu \Omega^{\mu\nu} \\ \frac{\partial \mathscr{L}_\omega}{\partial \omega_\nu} &= \frac{1}{2}m_\omega^2 (\delta_\mu^\nu \omega_\alpha g^{\alpha\mu} + \delta_\alpha^\nu \omega_\mu g^{\alpha\mu}) = m_\omega^2 \omega^\nu \end{aligned}$$

- Contributions also comes from $\mathscr{L}_{int}$.

$$\frac{\partial \mathscr{L}_{int}}{\partial \omega_\nu} = \frac{\partial \left(-\mathrm{g}_\omega \overline{\psi} \gamma^\mu \psi \omega_\mu \right)}{\partial \omega_\nu} = -\mathrm{g}_\omega \overline{\psi} \gamma^\nu \psi$$

Euler-Lagrange Equations

ω_ν field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \omega_\nu)} \right) = \frac{\partial \mathscr{L}}{\partial \omega_\nu}$

- The free Lagrangian is given by

$$\mathscr{L}_\omega = -\frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2 \omega_\mu \omega^\mu = -\frac{1}{4}\Omega_{\mu\nu}\Omega^{\mu\nu} + \frac{1}{2}m_\omega^2 \omega_\mu \omega_\alpha g^{\alpha\mu}$$

- Following exactly the same technique of field strength tensor $F_{\mu\nu}$ of Lec-3, one can write

$$\begin{aligned} \partial_\mu \left(\frac{\partial \mathscr{L}_\omega}{\partial(\partial_\mu \omega_\nu)} \right) &= -\partial_\mu \Omega^{\mu\nu} \\ \frac{\partial \mathscr{L}_\omega}{\partial \omega_\nu} &= \frac{1}{2}m_\omega^2 (\delta_\mu^\nu \omega_\alpha g^{\alpha\mu} + \delta_\alpha^\nu \omega_\mu g^{\alpha\mu}) = m_\omega^2 \omega^\nu \end{aligned}$$

- Contributions also comes from $\mathscr{L}_{int}$.

$$\frac{\partial \mathscr{L}_{int}}{\partial \omega_\nu} = \frac{\partial \left(-\mathrm{g}_\omega \overline{\psi} \gamma^\mu \psi \omega_\mu \right)}{\partial \omega_\nu} = -\mathrm{g}_\omega \overline{\psi} \gamma^\nu \psi \Rightarrow \boxed{\partial_\mu \Omega^{\mu\nu} + m_\omega^2 \omega^\nu = \mathrm{g}_\omega \overline{\psi} \gamma^\nu \psi}.$$

Euler-Lagrange Equations

$\overline{\psi}$ field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial \left(\partial_\mu \overline{\psi} \right)} \right) = \frac{\partial \mathscr{L}}{\partial \overline{\psi}}$

Euler-Lagrange Equations

$\overline{\psi}$ field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \overline{\psi})} \right) = \frac{\partial \mathscr{L}}{\partial \overline{\psi}}$

- The free Lagrangian is given by

$$\mathscr{L}_\psi = \overline{\psi}(i\gamma^\mu \partial_\mu - m)\psi.$$

Euler-Lagrange Equations

$\overline{\psi}$ field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \overline{\psi})} \right) = \frac{\partial \mathscr{L}}{\partial \overline{\psi}}$

- The free Lagrangian is given by

$$\mathscr{L}_\psi = \overline{\psi}(i\gamma^\mu \partial_\mu - m)\psi.$$

- The E-L equation is the Dirac equation

$$(i\gamma^\mu \partial_\mu - m)\psi = 0.$$

Euler-Lagrange Equations

$\overline{\psi}$ field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \overline{\psi})} \right) = \frac{\partial \mathscr{L}}{\partial \overline{\psi}}$

- The free Lagrangian is given by

$$\mathscr{L}_\psi = \overline{\psi}(i\gamma^\mu \partial_\mu - m)\psi.$$

- The E-L equation is the Dirac equation

$$(i\gamma^\mu \partial_\mu - m)\psi = 0.$$

- Contributions also comes from $\mathscr{L}_{int}$.

Euler-Lagrange Equations

$\overline{\psi}$ field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \overline{\psi})} \right) = \frac{\partial \mathscr{L}}{\partial \overline{\psi}}$

- The free Lagrangian is given by

$$\mathscr{L}_\psi = \overline{\psi}(i\gamma^\mu \partial_\mu - m)\psi.$$

- The E-L equation is the Dirac equation

$$(i\gamma^\mu \partial_\mu - m)\psi = 0.$$

- Contributions also comes from $\mathscr{L}_{int}$.

$$\frac{\partial \mathscr{L}_{int}}{\partial \overline{\psi}} = \frac{\partial \left(\mathrm{g}_\sigma \overline{\psi}\psi\sigma - \mathrm{g}_\omega \overline{\psi}\gamma^\mu \psi \omega_\mu \right)}{\partial \overline{\psi}}$$

Euler-Lagrange Equations

$\overline{\psi}$ field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \overline{\psi})} \right) = \frac{\partial \mathscr{L}}{\partial \overline{\psi}}$

- The free Lagrangian is given by

$$\mathscr{L}_\psi = \overline{\psi}(i\gamma^\mu \partial_\mu - m)\psi.$$

- The E-L equation is the Dirac equation

$$(i\gamma^\mu \partial_\mu - m)\psi = 0.$$

- Contributions also comes from $\mathscr{L}_{int}$.

$$\frac{\partial \mathscr{L}_{int}}{\partial \overline{\psi}} = \frac{\partial \left(\mathrm{g}_\sigma \overline{\psi}\psi\sigma - \mathrm{g}_\omega \overline{\psi}\gamma^\mu \psi \omega_\mu \right)}{\partial \overline{\psi}} = \mathrm{g}_\sigma \psi\sigma - \mathrm{g}_\omega \gamma^\mu \psi \omega_\mu$$

Euler-Lagrange Equations

$\overline{\psi}$ field $\rightarrow \partial_\mu \left(\frac{\partial \mathscr{L}}{\partial(\partial_\mu \overline{\psi})} \right) = \frac{\partial \mathscr{L}}{\partial \overline{\psi}}$

- The free Lagrangian is given by

$$\mathscr{L}_\psi = \overline{\psi}(i\gamma^\mu \partial_\mu - m)\psi.$$

- The E-L equation is the Dirac equation

$$(i\gamma^\mu \partial_\mu - m)\psi = 0.$$

- Contributions also comes from $\mathscr{L}_{int}$.

$$\frac{\partial \mathscr{L}_{int}}{\partial \overline{\psi}} = \frac{\partial \left(\mathrm{g}_\sigma \overline{\psi}\psi\sigma - \mathrm{g}_\omega \overline{\psi}\gamma^\mu \psi\omega_\mu \right)}{\partial \overline{\psi}} = \mathrm{g}_\sigma \psi\sigma - \mathrm{g}_\omega \gamma^\mu \psi\omega_\mu$$

- Combining everything together the equation becomes

$$\boxed{\left[\gamma^\mu \left(i\partial_\mu - \mathrm{g}_\omega \omega_\mu\right) - \left(m - \mathrm{g}_\sigma \sigma\right)\right]\psi = 0}.$$

Simplification of Euler-Lagrange Equations

Invoking symmetries

Simplification of Euler-Lagrange Equations

Invoking symmetries

- For the ω^ν field

$$\partial_\mu \Omega^{\mu\nu} + m_\omega^2 \omega^\nu = g_\omega \overline{\psi} \gamma^\nu \psi$$

Simplification of Euler-Lagrange Equations

Invoking symmetries

- For the ω^ν field

$$\partial_\mu \Omega^{\mu\nu} + m_\omega^2 \omega^\nu = g_\omega \overline{\psi} \gamma^\nu \psi = g_\omega j^\nu ,$$

Simplification of Euler-Lagrange Equations

Invoking symmetries

- For the ω^ν field

$$\partial_\mu \Omega^{\mu\nu} + m_\omega^2 \omega^\nu = \mathrm{g}_\omega \overline{\psi} \gamma^\nu \psi = \mathrm{g}_\omega j^\nu,$$

where, j^ν is the Noether current corresponding to the transformation $\psi \to e^{-iq\theta}\psi$.

Simplification of Euler-Lagrange Equations

Invoking symmetries

- For the ω^ν field

$$\partial_\mu \Omega^{\mu\nu} + m_\omega^2 \omega^\nu = g_\omega \overline{\psi} \gamma^\nu \psi = g_\omega j^\nu,$$

where, j^ν is the Noether current corresponding to the transformation $\psi \to e^{-iq\theta}\psi$. This is so called "*charge conservation*".

Simplification of Euler-Lagrange Equations

Invoking symmetries

- For the ω^ν field

$$\partial_\mu \Omega^{\mu\nu} + m_\omega^2 \omega^\nu = g_\omega \overline{\psi} \gamma^\nu \psi = g_\omega j^\nu,$$

  where, j^ν is the Noether current corresponding to the transformation $\psi \to e^{-iq\theta}\psi$. This is so called "*charge conservation*".
- We can write now,

$$\partial_\nu \partial_\mu \Omega^{\mu\nu} + m_\omega^2 \partial_\nu \omega^\nu = g_\omega \partial_\nu j^\nu$$

Simplification of Euler-Lagrange Equations

Invoking symmetries

- For the ω^ν field

$$\partial_\mu \Omega^{\mu\nu} + m_\omega^2 \omega^\nu = g_\omega \overline{\psi} \gamma^\nu \psi = g_\omega j^\nu,$$

  where, j^ν is the Noether current corresponding to the transformation $\psi \to e^{-iq\theta}\psi$. This is so called "*charge conservation*".
- We can write now,

$$\begin{aligned} & \partial_\nu \partial_\mu \Omega^{\mu\nu} + m_\omega^2 \partial_\nu \omega^\nu = g_\omega \partial_\nu j^\nu \\ \Rightarrow \quad & \partial_\mu \partial_\nu \Omega^{\mu\nu} + m_\omega^2 \partial_\nu \omega^\nu = 0 \end{aligned}$$

Simplification of Euler-Lagrange Equations

Invoking symmetries

- For the ω^ν field

$$\partial_\mu \Omega^{\mu\nu} + m_\omega^2 \omega^\nu = g_\omega \overline{\psi} \gamma^\nu \psi = g_\omega j^\nu,$$

where, j^ν is the Noether current corresponding to the transformation $\psi \to e^{-iq\theta}\psi$. This is so called "*charge conservation*".

- We can write now,

$$\begin{aligned} & \partial_\nu \partial_\mu \Omega^{\mu\nu} + m_\omega^2 \partial_\nu \omega^\nu = g_\omega \partial_\nu j^\nu \\ \Rightarrow \quad & \partial_\mu \partial_\nu \Omega^{\mu\nu} + m_\omega^2 \partial_\nu \omega^\nu = 0 \\ \Rightarrow \quad & -\partial_\mu \partial_\nu \Omega^{\nu\mu} + m_\omega^2 \partial_\nu \omega^\nu = 0 \end{aligned}$$

Simplification of Euler-Lagrange Equations

Invoking symmetries

- For the ω^ν field

$$\partial_\mu \Omega^{\mu\nu} + m_\omega^2 \omega^\nu = g_\omega \overline{\psi} \gamma^\nu \psi = g_\omega j^\nu,$$

where, j^ν is the Noether current corresponding to the transformation $\psi \to e^{-iq\theta}\psi$. This is so called "*charge conservation*".

- We can write now,

$$\begin{aligned} & \partial_\nu \partial_\mu \Omega^{\mu\nu} + m_\omega^2 \partial_\nu \omega^\nu = g_\omega \partial_\nu j^\nu \\ \Rightarrow \quad & \partial_\mu \partial_\nu \Omega^{\mu\nu} + m_\omega^2 \partial_\nu \omega^\nu = 0 \\ \Rightarrow \quad & -\partial_\mu \partial_\nu \Omega^{\nu\mu} + m_\omega^2 \partial_\nu \omega^\nu = 0 \\ \Rightarrow \quad & \boxed{\partial_\nu \omega^\nu = 0} \end{aligned}$$

Simplification of Euler-Lagrange Equations

Invoking symmetries

- For the ω^ν field

$$\partial_\mu \Omega^{\mu\nu} + m_\omega^2 \omega^\nu = g_\omega \overline{\psi} \gamma^\nu \psi = g_\omega j^\nu,$$

where, j^ν is the Noether current corresponding to the transformation $\psi \to e^{-iq\theta}\psi$. This is so called "*charge conservation*".

- We can write now,

$$\begin{aligned}
& \partial_\nu \partial_\mu \Omega^{\mu\nu} + m_\omega^2 \partial_\nu \omega^\nu = g_\omega \partial_\nu j^\nu \\
\Rightarrow \quad & \partial_\mu \partial_\nu \Omega^{\mu\nu} + m_\omega^2 \partial_\nu \omega^\nu = 0 \\
\Rightarrow \quad & -\partial_\mu \partial_\nu \Omega^{\nu\mu} + m_\omega^2 \partial_\nu \omega^\nu = 0 \\
\Rightarrow \quad & \boxed{\partial_\nu \omega^\nu = 0} \\
\Rightarrow \quad & \partial_\mu \Omega^{\mu\nu} + m_\omega^2 \omega^\nu
\end{aligned}$$

Simplification of Euler-Lagrange Equations

Invoking symmetries

- For the ω^ν field

$$\partial_\mu \Omega^{\mu\nu} + m_\omega^2 \omega^\nu = g_\omega \overline{\psi} \gamma^\nu \psi = g_\omega j^\nu,$$

where, j^ν is the Noether current corresponding to the transformation $\psi \to e^{-iq\theta}\psi$. This is so called "*charge conservation*".

- We can write now,

$$\begin{aligned}
& \partial_\nu \partial_\mu \Omega^{\mu\nu} + m_\omega^2 \partial_\nu \omega^\nu = g_\omega \partial_\nu j^\nu \\
\Rightarrow \quad & \partial_\mu \partial_\nu \Omega^{\mu\nu} + m_\omega^2 \partial_\nu \omega^\nu = 0 \\
\Rightarrow \quad & -\partial_\mu \partial_\nu \Omega^{\nu\mu} + m_\omega^2 \partial_\nu \omega^\nu = 0 \\
\Rightarrow \quad & \boxed{\partial_\nu \omega^\nu = 0} \\
\Rightarrow \quad & \partial_\mu \Omega^{\mu\nu} + m_\omega^2 \omega^\nu = \partial_\mu (\partial^\mu \omega^\nu - \partial^\nu \omega^\mu) + m_\omega^2 \omega^\nu
\end{aligned}$$

Simplification of Euler-Lagrange Equations

Invoking symmetries

- For the ω^ν field

$$\partial_\mu \Omega^{\mu\nu} + m_\omega^2 \omega^\nu = g_\omega \overline{\psi} \gamma^\nu \psi = g_\omega j^\nu,$$

where, j^ν is the Noether current corresponding to the transformation $\psi \to e^{-iq\theta}\psi$. This is so called "*charge conservation*".

- We can write now,

$$\begin{aligned}
& \partial_\nu \partial_\mu \Omega^{\mu\nu} + m_\omega^2 \partial_\nu \omega^\nu = g_\omega \partial_\nu j^\nu \\
\Rightarrow \quad & \partial_\mu \partial_\nu \Omega^{\mu\nu} + m_\omega^2 \partial_\nu \omega^\nu = 0 \\
\Rightarrow \quad & -\partial_\mu \partial_\nu \Omega^{\nu\mu} + m_\omega^2 \partial_\nu \omega^\nu = 0 \\
\Rightarrow \quad & \boxed{\partial_\nu \omega^\nu = 0} \\
\Rightarrow \quad & \partial_\mu \Omega^{\mu\nu} + m_\omega^2 \omega^\nu = \partial_\mu (\partial^\mu \omega^\nu - \partial^\nu \omega^\mu) + m_\omega^2 \omega^\nu \\
& = \partial_\mu \partial^\mu \omega^\nu - \partial^\nu \partial_\mu \omega^\mu + m_\omega^2 \omega^\nu
\end{aligned}$$

Simplification of Euler-Lagrange Equations

Invoking symmetries

- For the ω^ν field

$$\partial_\mu \Omega^{\mu\nu} + m_\omega^2 \omega^\nu = g_\omega \overline{\psi} \gamma^\nu \psi = g_\omega j^\nu,$$

where, j^ν is the Noether current corresponding to the transformation $\psi \to e^{-iq\theta}\psi$. This is so called "*charge conservation*".

- We can write now,

$$\begin{aligned}
& \partial_\nu \partial_\mu \Omega^{\mu\nu} + m_\omega^2 \partial_\nu \omega^\nu = g_\omega \partial_\nu j^\nu \\
\Rightarrow \quad & \partial_\mu \partial_\nu \Omega^{\mu\nu} + m_\omega^2 \partial_\nu \omega^\nu = 0 \\
\Rightarrow \quad & -\partial_\mu \partial_\nu \Omega^{\nu\mu} + m_\omega^2 \partial_\nu \omega^\nu = 0 \\
\Rightarrow \quad & \boxed{\partial_\nu \omega^\nu = 0} \\
\Rightarrow \quad & \partial_\mu \Omega^{\mu\nu} + m_\omega^2 \omega^\nu = \partial_\mu (\partial^\mu \omega^\nu - \partial^\nu \omega^\mu) + m_\omega^2 \omega^\nu \\
& = \partial_\mu \partial^\mu \omega^\nu - \partial^\nu \partial_\mu \omega^\mu + m_\omega^2 \omega^\nu = \partial_\mu \partial^\mu \omega^\nu + m_\omega^2 \omega^\nu
\end{aligned}$$

Simplification of Euler-Lagrange Equations

Invoking symmetries

- For the ω^ν field

$$\partial_\mu \Omega^{\mu\nu} + m_\omega^2 \omega^\nu = g_\omega \overline{\psi} \gamma^\nu \psi = g_\omega j^\nu,$$

where, j^ν is the Noether current corresponding to the transformation $\psi \to e^{-iq\theta}\psi$. This is so called "*charge conservation*".

- We can write now,

$$\begin{aligned}
& \partial_\nu \partial_\mu \Omega^{\mu\nu} + m_\omega^2 \partial_\nu \omega^\nu = g_\omega \partial_\nu j^\nu \\
\Rightarrow \quad & \partial_\mu \partial_\nu \Omega^{\mu\nu} + m_\omega^2 \partial_\nu \omega^\nu = 0 \\
\Rightarrow \quad & -\partial_\mu \partial_\nu \Omega^{\nu\mu} + m_\omega^2 \partial_\nu \omega^\nu = 0 \\
\Rightarrow \quad & \boxed{\partial_\nu \omega^\nu = 0} \\
\Rightarrow \quad & \partial_\mu \Omega^{\mu\nu} + m_\omega^2 \omega^\nu = \partial_\mu (\partial^\mu \omega^\nu - \partial^\nu \omega^\mu) + m_\omega^2 \omega^\nu \\
& = \partial_\mu \partial^\mu \omega^\nu - \partial^\nu \partial_\mu \omega^\mu + m_\omega^2 \omega^\nu = \partial_\mu \partial^\mu \omega^\nu + m_\omega^2 \omega^\nu \\
\Rightarrow \quad & \boxed{(\Box + m_\omega^2)\omega^\nu = g_\omega j^\nu}.
\end{aligned}$$

Simplification of Euler-Lagrange Equations

Invoking symmetries

Simplification of Euler-Lagrange Equations

Invoking symmetries

- **Problem 1: Show that under time reversal $\psi \to \widetilde{\psi} = T\psi^*$ the space components of the current j^μ vanish, where, $T = i\gamma^1\gamma^3$ and $j^\mu = \overline{\psi}\gamma^\mu\psi$.**

Simplification of Euler-Lagrange Equations

Invoking symmetries

- **Problem 1: Show that under time reversal $\psi \to \widetilde{\psi} = T\psi^*$ the space components of the current j^μ vanish, where, $T = i\gamma^1\gamma^3$ and $j^\mu = \overline{\psi}\gamma^\mu\psi$.**
- This simplifies further the field equation of ω^ν as

$$(\Box + m_\omega^2)\omega^0 = g_\omega j^0$$

Simplification of Euler-Lagrange Equations

Invoking symmetries

- **Problem 1: Show that under time reversal $\psi \to \widetilde{\psi} = T\psi^*$ the space components of the current j^μ vanish, where, $T = i\gamma^1\gamma^3$ and $j^\mu = \overline{\psi}\gamma^\mu\psi$.**
- This simplifies further the field equation of ω^ν as

$$(\Box + m_\omega^2)\omega^0 = g_\omega j^0 = g_\omega \psi^\dagger \psi$$

Simplification of Euler-Lagrange Equations

Invoking symmetries

- **Problem 1: Show that under time reversal $\psi \to \widetilde{\psi} = T\psi^*$ the space components of the current j^μ vanish, where, $T = i\gamma^1\gamma^3$ and $j^\mu = \overline{\psi}\gamma^\mu\psi$.**
- This simplifies further the field equation of ω^ν as

$$(\Box + m_\omega^2)\omega^0 = g_\omega j^0 = g_\omega \psi^\dagger \psi$$

$$\Rightarrow \quad \boxed{(-\nabla^2 + m_\omega^2)\omega^0 = g_\omega \rho_b}$$

under the static field approximation, where, $\psi^\dagger\psi = \rho_b$ is the nucleon density.

Simplification of Euler-Lagrange Equations

Invoking symmetries

- **Problem 1: Show that under time reversal $\psi \to \widetilde{\psi} = T\psi^*$ the space components of the current j^μ vanish, where, $T = i\gamma^1\gamma^3$ and $j^\mu = \overline{\psi}\gamma^\mu\psi$.**
- This simplifies further the field equation of ω^ν as

$$(\Box + m_\omega^2)\omega^0 = g_\omega j^0 = g_\omega \psi^\dagger \psi$$

$$\Rightarrow \quad \boxed{(-\nabla^2 + m_\omega^2)\omega^0 = g_\omega \rho_b}$$

  under the static field approximation, where, $\psi^\dagger\psi = \rho_b$ is the nucleon density.
- Similarly, under static field approximation, field equation for σ becomes

$$(\Box + m_\sigma^2)\sigma = g_\sigma \overline{\psi}\psi$$

Simplification of Euler-Lagrange Equations

Invoking symmetries

- **Problem 1: Show that under time reversal $\psi \to \widetilde{\psi} = T\psi^*$ the space components of the current j^μ vanish, where, $T = i\gamma^1\gamma^3$ and $j^\mu = \overline{\psi}\gamma^\mu\psi$.**
- This simplifies further the field equation of ω^ν as

$$(\Box + m_\omega^2)\omega^0 = g_\omega j^0 = g_\omega \psi^\dagger \psi$$

$$\Rightarrow \quad \boxed{(-\nabla^2 + m_\omega^2)\omega^0 = g_\omega \rho_b}$$

  under the static field approximation, where, $\psi^\dagger\psi = \rho_b$ is the nucleon density.
- Similarly, under static field approximation, field equation for σ becomes

$$(\Box + m_\sigma^2)\sigma = g_\sigma \overline{\psi}\psi = g_\sigma \psi^\dagger \gamma^0 \psi$$

Simplification of Euler-Lagrange Equations

Invoking symmetries

- **Problem 1: Show that under time reversal $\psi \to \widetilde{\psi} = T\psi^*$ the space components of the current j^μ vanish, where, $T = i\gamma^1\gamma^3$ and $j^\mu = \overline{\psi}\gamma^\mu\psi$.**
- This simplifies further the field equation of ω^ν as

$$(\Box + m_\omega^2)\omega^0 = g_\omega j^0 = g_\omega \psi^\dagger \psi$$

$$\Rightarrow \quad \boxed{(-\nabla^2 + m_\omega^2)\omega^0 = g_\omega \rho_b}$$

under the static field approximation, where, $\psi^\dagger\psi = \rho_b$ is the nucleon density.

- Similarly, under static field approximation, field equation for σ becomes

$$(\Box + m_\sigma^2)\sigma = g_\sigma \overline{\psi}\psi = g_\sigma \psi^\dagger \gamma^0 \psi$$

$$\Rightarrow \quad \boxed{(-\nabla^2 + m_\sigma^2)\sigma = g_\sigma \rho_s},$$

where, ρ_s is called as scalar density.

Dirac field in interaction

Mean Field formalism

Dirac field in interaction

Mean Field formalism

- We start again from the Dirac equation

$$\left[\gamma^{\mu}\left(i\partial_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}\right) - \left(m - \mathrm{g}_{\sigma}\sigma\right)\right]\psi(x) = 0$$

Dirac field in interaction

Mean Field formalism

- We start again from the Dirac equation

$$\left[\gamma^{\mu}\left(i\partial_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}\right) - \left(m - \mathrm{g}_{\sigma}\sigma\right)\right]\psi(x) = 0$$

$$\psi(x) = \psi(p)e^{-ip\cdot x}$$

Dirac field in interaction

Mean Field formalism

- We start again from the Dirac equation

$$\left[\gamma^{\mu}\left(i\partial_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}\right) - \left(m - \mathrm{g}_{\sigma}\sigma\right)\right]\psi(x) = 0$$

$$\psi(x) = \psi(p)e^{-ip\cdot x}$$

$$\Rightarrow \quad \left[\gamma^{\mu}\left(p_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}\right) - \left(m - \mathrm{g}_{\sigma}\sigma\right)\right]\psi(p) = 0$$

Dirac field in interaction

Mean Field formalism

- We start again from the Dirac equation

$$\left[\gamma^{\mu}\left(i\partial_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}\right) - \left(m - \mathrm{g}_{\sigma}\sigma\right)\right]\psi(x) = 0$$

$$\psi(x) = \psi(p)e^{-ip\cdot x}$$

$$\Rightarrow \quad \left[\gamma^{\mu}\left(p_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}\right) - \left(m - \mathrm{g}_{\sigma}\sigma\right)\right]\psi(p) = 0$$

$$k_{\mu} = p_{\mu} - \mathrm{g}_{\omega}\omega_{\mu},$$

Dirac field in interaction

Mean Field formalism

- We start again from the Dirac equation

$$\left[\gamma^{\mu}\left(i\partial_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}\right) - \left(m - \mathrm{g}_{\sigma}\sigma\right)\right]\psi(x) = 0$$

$$\psi(x) = \psi(p)e^{-ip\cdot x}$$

$$\Rightarrow \quad \left[\gamma^{\mu}\left(p_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}\right) - \left(m - \mathrm{g}_{\sigma}\sigma\right)\right]\psi(p) = 0$$

$$k_{\mu} = p_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}, \;\; m^{*} = m - \mathrm{g}_{\sigma}\sigma$$

Dirac field in interaction

Mean Field formalism

- We start again from the Dirac equation

$$\left[\gamma^{\mu}\left(i\partial_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}\right) - \left(m - \mathrm{g}_{\sigma}\sigma\right)\right]\psi(x) = 0$$

$$\psi(x) = \psi(p)e^{-ip\cdot x}$$

$$\Rightarrow \quad \left[\gamma^{\mu}\left(p_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}\right) - \left(m - \mathrm{g}_{\sigma}\sigma\right)\right]\psi(p) = 0$$

$$k_{\mu} = p_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}, \;\; m^{*} = m - \mathrm{g}_{\sigma}\sigma$$

$$\Rightarrow \quad (\gamma^{\mu}k_{\mu} - m^{*})\psi(k) = 0$$

Dirac field in interaction

Mean Field formalism

- We start again from the Dirac equation

$$\left[\gamma^{\mu}\left(i\partial_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}\right) - \left(m - \mathrm{g}_{\sigma}\sigma\right)\right]\psi(x) = 0$$

$$\psi(x) = \psi(p)e^{-ip\cdot x}$$

$$\Rightarrow \quad \left[\gamma^{\mu}\left(p_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}\right) - \left(m - \mathrm{g}_{\sigma}\sigma\right)\right]\psi(p) = 0$$

$$k_{\mu} = p_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}, \;\; m^{*} = m - \mathrm{g}_{\sigma}\sigma$$

$$\Rightarrow \quad (\gamma^{\mu}k_{\mu} + m^{*})(\gamma^{\mu}k_{\mu} - m^{*})\psi(k) = 0$$

Dirac field in interaction

Mean Field formalism

- We start again from the Dirac equation

$$\left[\gamma^\mu\left(i\partial_\mu - g_\omega\omega_\mu\right) - \left(m - g_\sigma\sigma\right)\right]\psi(x) = 0$$

$$\psi(x) = \psi(p)e^{-ip\cdot x}$$

$$\Rightarrow \quad \left[\gamma^\mu\left(p_\mu - g_\omega\omega_\mu\right) - \left(m - g_\sigma\sigma\right)\right]\psi(p) = 0$$

$$k_\mu = p_\mu - g_\omega\omega_\mu, \;\; m^* = m - g_\sigma\sigma$$

$$\Rightarrow \quad (\gamma^\mu k_\mu + m^*)(\gamma^\mu k_\mu - m^*)\psi(k) = 0$$

$$\Rightarrow \quad (k_\mu k^\mu - {m^*}^2)\psi(k) = 0$$

Dirac field in interaction

Mean Field formalism

- We start again from the Dirac equation

$$\left[\gamma^\mu\left(i\partial_\mu - g_\omega\omega_\mu\right) - \left(m - g_\sigma\sigma\right)\right]\psi(x) = 0$$

$$\psi(x) = \psi(p)e^{-ip\cdot x}$$

$$\Rightarrow \quad \left[\gamma^\mu\left(p_\mu - g_\omega\omega_\mu\right) - \left(m - g_\sigma\sigma\right)\right]\psi(p) = 0$$

$$k_\mu = p_\mu - g_\omega\omega_\mu, \;\; m^* = m - g_\sigma\sigma$$

$$\Rightarrow \quad (\gamma^\mu k_\mu + m^*)(\gamma^\mu k_\mu - m^*)\psi(k) = 0$$

$$\Rightarrow \quad (k_\mu k^\mu - m^{*2})\psi(k) = 0$$

$$\Rightarrow \quad k_0 = \sqrt{\boldsymbol{k}^2 + m^{*2}}$$

Dirac field in interaction

Mean Field formalism

- We start again from the Dirac equation

$$\left[\gamma^\mu\left(i\partial_\mu - \mathrm{g}_\omega\omega_\mu\right) - \left(m - \mathrm{g}_\sigma\sigma\right)\right]\psi(x) = 0$$

$$\psi(x) = \psi(p)e^{-ip\cdot x}$$

$$\Rightarrow \quad \left[\gamma^\mu\left(p_\mu - \mathrm{g}_\omega\omega_\mu\right) - \left(m - \mathrm{g}_\sigma\sigma\right)\right]\psi(p) = 0$$

$$k_\mu = p_\mu - \mathrm{g}_\omega\omega_\mu, \;\; m^* = m - \mathrm{g}_\sigma\sigma$$

$$\Rightarrow \quad (\gamma^\mu k_\mu + m^*)(\gamma^\mu k_\mu - m^*)\psi(k) = 0$$

$$\Rightarrow \quad (k_\mu k^\mu - m^{*2})\psi(k) = 0$$

$$\Rightarrow \quad k_0 = \sqrt{\boldsymbol{k}^2 + m^{*2}} \Rightarrow e(\boldsymbol{p}) \equiv p_0(\boldsymbol{p}) = k_0 + \mathrm{g}_\omega\omega_0$$

Dirac field in interaction

Mean Field formalism

- We start again from the Dirac equation

$$\left[\gamma^\mu\left(i\partial_\mu - g_\omega\omega_\mu\right) - \left(m - g_\sigma\sigma\right)\right]\psi(x) = 0$$

$$\psi(x) = \psi(p)e^{-ip\cdot x}$$

$$\Rightarrow \quad \left[\gamma^\mu\left(p_\mu - g_\omega\omega_\mu\right) - \left(m - g_\sigma\sigma\right)\right]\psi(p) = 0$$

$$k_\mu = p_\mu - g_\omega\omega_\mu, \;\; m^* = m - g_\sigma\sigma$$

$$\Rightarrow \quad (\gamma^\mu k_\mu + m^*)(\gamma^\mu k_\mu - m^*)\psi(k) = 0$$

$$\Rightarrow \quad (k_\mu k^\mu - m^{*2})\psi(k) = 0$$

$$\Rightarrow \quad k_0 = \sqrt{\boldsymbol{k}^2 + m^{*2}} \Rightarrow e(\boldsymbol{p}) \equiv p_0(\boldsymbol{p}) = k_0 + g_\omega\omega_0$$

$$\text{where, } k_0 = E(\boldsymbol{p}) = \sqrt{(\boldsymbol{p} - g_\omega\boldsymbol{\omega})^2 + (m - g_\sigma\sigma)^2}$$

Dirac field in interaction

Mean Field formalism

- We start again from the Dirac equation

$$\left[\gamma^\mu\left(i\partial_\mu - g_\omega\omega_\mu\right) - \left(m - g_\sigma\sigma\right)\right]\psi(x) = 0$$

$$\psi(x) = \psi(p)e^{-ip\cdot x}$$

$$\Rightarrow \quad \left[\gamma^\mu\left(p_\mu - g_\omega\omega_\mu\right) - \left(m - g_\sigma\sigma\right)\right]\psi(p) = 0$$

$$k_\mu = p_\mu - g_\omega\omega_\mu, \;\; m^* = m - g_\sigma\sigma$$

$$\Rightarrow \quad (\gamma^\mu k_\mu + m^*)(\gamma^\mu k_\mu - m^*)\psi(k) = 0$$

$$\Rightarrow \quad (k_\mu k^\mu - m^{*2})\psi(k) = 0$$

$$\Rightarrow \quad k_0 = \sqrt{\boldsymbol{k}^2 + m^{*2}} \Rightarrow e(\boldsymbol{p}) \equiv p_0(\boldsymbol{p}) = k_0 + g_\omega\omega_0$$

$$\text{where, } k_0 = E(\boldsymbol{p}) = \sqrt{(\boldsymbol{p} - g_\omega\boldsymbol{\omega})^2 + (m - g_\sigma\sigma)^2}$$

$$= \sqrt{\boldsymbol{p}^2 + (m - g_\sigma\sigma)^2}$$

Dirac field in interaction

Mean Field formalism

- We start again from the Dirac equation

$$\left[\gamma^\mu\left(i\partial_\mu - g_\omega\omega_\mu\right) - \left(m - g_\sigma\sigma\right)\right]\psi(x) = 0$$

$$\psi(x) = \psi(p)e^{-ip\cdot x}$$

$$\Rightarrow \quad \left[\gamma^\mu\left(p_\mu - g_\omega\omega_\mu\right) - \left(m - g_\sigma\sigma\right)\right]\psi(p) = 0$$

$$k_\mu = p_\mu - g_\omega\omega_\mu, \;\; m^* = m - g_\sigma\sigma$$

$$\Rightarrow \quad (\gamma^\mu k_\mu + m^*)(\gamma^\mu k_\mu - m^*)\psi(k) = 0$$

$$\Rightarrow \quad (k_\mu k^\mu - m^{*2})\psi(k) = 0$$

$$\Rightarrow \quad k_0 = \sqrt{\boldsymbol{k}^2 + m^{*2}} \Rightarrow e(\boldsymbol{p}) \equiv p_0(\boldsymbol{p}) = k_0 + g_\omega\omega_0$$

$$\text{where, } k_0 = E(\boldsymbol{p}) = \sqrt{(\boldsymbol{p} - g_\omega\boldsymbol{\omega})^2 + (m - g_\sigma\sigma)^2}$$

$$= \sqrt{\boldsymbol{p}^2 + (m - g_\sigma\sigma)^2}$$

- **Problem 2: Show that $(\gamma^\mu k_\mu + m)(\gamma^\mu k_\mu - m) = (k_\mu k^\mu - m^2)$.**

Simplification of Euler-Lagrange Equations

Dirac equation in Pauli-matrix notation

Simplification of Euler-Lagrange Equations

Dirac equation in Pauli-matrix notation

- We start from the Dirac equation in interaction,

$$\left[\gamma^{\mu}\left(i\partial_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}\right) - \left(m - \mathrm{g}_{\sigma}\sigma\right)\right]\psi = 0$$

Simplification of Euler-Lagrange Equations

Dirac equation in Pauli-matrix notation

- We start from the Dirac equation in interaction,

$$\left[\gamma^{\mu}\left(i\partial_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}\right) - \left(m - \mathrm{g}_{\sigma}\sigma\right)\right]\psi = 0$$

$$\Rightarrow \quad \left[\gamma^{0}\left(i\partial_{0} - \mathrm{g}_{\omega}\omega_{0}\right) - \boldsymbol{\gamma}\cdot\left(\boldsymbol{p} - \mathrm{g}_{\omega}\boldsymbol{\omega}\right) - m^{*}\right]\psi = 0$$

Simplification of Euler-Lagrange Equations

Dirac equation in Pauli-matrix notation

- We start from the Dirac equation in interaction,

$$\left[\gamma^{\mu}\left(i\partial_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}\right) - \left(m - \mathrm{g}_{\sigma}\sigma\right)\right]\psi = 0$$

$$\Rightarrow \quad \left[\gamma^{0}\left(i\partial_{0} - \mathrm{g}_{\omega}\omega_{0}\right) - \boldsymbol{\gamma}\cdot\left(\boldsymbol{p} - \mathrm{g}_{\omega}\boldsymbol{\omega}\right) - m^{*}\right]\psi = 0$$

$$\Rightarrow \quad \boxed{\left[\gamma_{0}\left\{\boldsymbol{\gamma}\cdot\boldsymbol{p} + m^{*}\right\} + \mathrm{g}_{\omega}\omega_{0}\right]\psi = H_{D}\psi}.$$

Simplification of Euler-Lagrange Equations

Dirac equation in Pauli-matrix notation

- We start from the Dirac equation in interaction,

$$\left[\gamma^{\mu}\left(i\partial_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}\right) - \left(m - \mathrm{g}_{\sigma}\sigma\right)\right]\psi = 0$$

$$\Rightarrow \quad \left[\gamma^{0}\left(i\partial_{0} - \mathrm{g}_{\omega}\omega_{0}\right) - \boldsymbol{\gamma}\cdot\left(\boldsymbol{p} - \mathrm{g}_{\omega}\boldsymbol{\omega}\right) - m^{*}\right]\psi = 0$$

$$\Rightarrow \quad \boxed{\left[\gamma_{0}\left\{\boldsymbol{\gamma}\cdot\boldsymbol{p} + m^{*}\right\} + \mathrm{g}_{\omega}\omega_{0}\right]\psi = H_{D}\psi}.$$

$$\Rightarrow \quad \left(\psi^{\dagger}H_{D}\psi\right)_{\boldsymbol{p}} = e(\boldsymbol{p}) = E(\boldsymbol{p}) + \mathrm{g}_{\omega}\omega_{0}$$

Simplification of Euler-Lagrange Equations

Dirac equation in Pauli-matrix notation

- We start from the Dirac equation in interaction,

$$
\begin{aligned}
& \left[\gamma^{\mu}\left(i\partial_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}\right) - \left(m - \mathrm{g}_{\sigma}\sigma\right)\right]\psi = 0 \\
\Rightarrow \quad & \left[\gamma^{0}\left(i\partial_{0} - \mathrm{g}_{\omega}\omega_{0}\right) - \boldsymbol{\gamma}\cdot\left(\boldsymbol{p} - \mathrm{g}_{\omega}\boldsymbol{\omega}\right) - m^{*}\right]\psi = 0 \\
\Rightarrow \quad & \boxed{\left[\gamma_{0}\left\{\boldsymbol{\gamma}\cdot\boldsymbol{p} + m^{*}\right\} + \mathrm{g}_{\omega}\omega_{0}\right]\psi = H_{D}\psi}. \\
\Rightarrow \quad & (\psi^{\dagger}H_{D}\psi)_{\boldsymbol{p}} = e(\boldsymbol{p}) = E(\boldsymbol{p}) + \mathrm{g}_{\omega}\omega_{0} \\
\Rightarrow \quad & \frac{\partial}{\partial\zeta}(\psi^{\dagger}H_{D}\psi)_{\boldsymbol{p}} = (\psi^{\dagger}\frac{\partial H_{D}}{\partial\zeta}\psi)_{\boldsymbol{p}} + e(\boldsymbol{p})\frac{\partial}{\partial\zeta}(\psi^{\dagger}\psi)_{\boldsymbol{p}}
\end{aligned}
$$

Simplification of Euler-Lagrange Equations

Dirac equation in Pauli-matrix notation

- We start from the Dirac equation in interaction,

$$
\begin{aligned}
& \left[\gamma^{\mu}\left(i\partial_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}\right) - \left(m - \mathrm{g}_{\sigma}\sigma\right)\right]\psi = 0 \\
\Rightarrow \quad & \left[\gamma^{0}\left(i\partial_{0} - \mathrm{g}_{\omega}\omega_{0}\right) - \boldsymbol{\gamma}\cdot\left(\boldsymbol{p} - \mathrm{g}_{\omega}\boldsymbol{\omega}\right) - m^{*}\right]\psi = 0 \\
\Rightarrow \quad & \boxed{\left[\gamma_{0}\left\{\boldsymbol{\gamma}\cdot\boldsymbol{p} + m^{*}\right\} + \mathrm{g}_{\omega}\omega_{0}\right]\psi = H_{D}\psi}. \\
\Rightarrow \quad & (\psi^{\dagger}H_{D}\psi)_{\boldsymbol{p}} = e(\boldsymbol{p}) = E(\boldsymbol{p}) + \mathrm{g}_{\omega}\omega_{0} \\
\Rightarrow \quad & \frac{\partial}{\partial\zeta}(\psi^{\dagger}H_{D}\psi)_{\boldsymbol{p}} = (\psi^{\dagger}\frac{\partial H_{D}}{\partial\zeta}\psi)_{\boldsymbol{p}} + e(\boldsymbol{p})\frac{\partial}{\partial\zeta}(\psi^{\dagger}\psi)_{\boldsymbol{p}}
\end{aligned}
$$

- Under mean-field approximation σ and ω^{μ} field equations can be further simplified as,

$$(-\nabla^{2} + m_{\sigma}^{2})\sigma$$

Simplification of Euler-Lagrange Equations

Dirac equation in Pauli-matrix notation

- We start from the Dirac equation in interaction,

$$
\begin{aligned}
& \left[\gamma^{\mu}\left(i\partial_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}\right) - \left(m - \mathrm{g}_{\sigma}\sigma\right)\right]\psi = 0 \\
\Rightarrow \quad & \left[\gamma^{0}\left(i\partial_{0} - \mathrm{g}_{\omega}\omega_{0}\right) - \boldsymbol{\gamma}\cdot\left(\boldsymbol{p} - \mathrm{g}_{\omega}\boldsymbol{\omega}\right) - m^{*}\right]\psi = 0 \\
\Rightarrow \quad & \boxed{\left[\gamma_{0}\left\{\boldsymbol{\gamma}\cdot\boldsymbol{p} + m^{*}\right\} + \mathrm{g}_{\omega}\omega_{0}\right]\psi = H_{D}\psi}. \\
\Rightarrow \quad & (\psi^{\dagger}H_{D}\psi)_{\boldsymbol{p}} = e(\boldsymbol{p}) = E(\boldsymbol{p}) + \mathrm{g}_{\omega}\omega_{0} \\
\Rightarrow \quad & \frac{\partial}{\partial\zeta}(\psi^{\dagger}H_{D}\psi)_{\boldsymbol{p}} = (\psi^{\dagger}\frac{\partial H_{D}}{\partial\zeta}\psi)_{\boldsymbol{p}} + e(\boldsymbol{p})\frac{\partial}{\partial\zeta}(\psi^{\dagger}\psi)_{\boldsymbol{p}}
\end{aligned}
$$

- Under mean-field approximation σ and ω^{μ} field equations can be further simplified as,

$$
(-\nabla^{2} + m_{\sigma}^{2})\sigma = m_{\sigma}^{2}\langle\sigma\rangle = \mathrm{g}_{\sigma}\langle\overline{\psi}\psi\rangle = \mathrm{g}_{\sigma}\rho_{s}
$$

Simplification of Euler-Lagrange Equations

Dirac equation in Pauli-matrix notation

- We start from the Dirac equation in interaction,

$$
\begin{aligned}
& \left[\gamma^{\mu}\left(i\partial_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}\right) - \left(m - \mathrm{g}_{\sigma}\sigma\right)\right]\psi = 0 \\
\Rightarrow \quad & \left[\gamma^{0}\left(i\partial_{0} - \mathrm{g}_{\omega}\omega_{0}\right) - \boldsymbol{\gamma}\cdot\left(\boldsymbol{p} - \mathrm{g}_{\omega}\boldsymbol{\omega}\right) - m^{*}\right]\psi = 0 \\
\Rightarrow \quad & \boxed{\left[\gamma_{0}\left\{\boldsymbol{\gamma}\cdot\boldsymbol{p} + m^{*}\right\} + \mathrm{g}_{\omega}\omega_{0}\right]\psi = H_{D}\psi}. \\
\Rightarrow \quad & (\psi^{\dagger}H_{D}\psi)_{\boldsymbol{p}} = e(\boldsymbol{p}) = E(\boldsymbol{p}) + \mathrm{g}_{\omega}\omega_{0} \\
\Rightarrow \quad & \frac{\partial}{\partial\zeta}(\psi^{\dagger}H_{D}\psi)_{\boldsymbol{p}} = (\psi^{\dagger}\frac{\partial H_{D}}{\partial\zeta}\psi)_{\boldsymbol{p}} + e(\boldsymbol{p})\frac{\partial}{\partial\zeta}(\psi^{\dagger}\psi)_{\boldsymbol{p}}
\end{aligned}
$$

- Under mean-field approximation σ and ω^{μ} field equations can be further simplified as,

$$
(-\nabla^{2} + m_{\sigma}^{2})\sigma = m_{\sigma}^{2}\langle\sigma\rangle = \mathrm{g}_{\sigma}\langle\overline{\psi}\psi\rangle = \mathrm{g}_{\sigma}\rho_{s} \Rightarrow \boxed{\mathrm{g}_{\sigma}\langle\sigma\rangle = \left(\frac{\mathrm{g}_{\sigma}}{m_{\sigma}}\right)^{2}\langle\overline{\psi}\psi\rangle}.
$$

Simplification of Euler-Lagrange Equations

Dirac equation in Pauli-matrix notation

- We start from the Dirac equation in interaction,

$$
\begin{aligned}
& \left[\gamma^{\mu}\left(i\partial_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}\right) - \left(m - \mathrm{g}_{\sigma}\sigma\right)\right]\psi = 0 \\
\Rightarrow \quad & \left[\gamma^{0}\left(i\partial_{0} - \mathrm{g}_{\omega}\omega_{0}\right) - \boldsymbol{\gamma}\cdot\left(\boldsymbol{p} - \mathrm{g}_{\omega}\boldsymbol{\omega}\right) - m^{*}\right]\psi = 0 \\
\Rightarrow \quad & \boxed{\left[\gamma_{0}\left\{\boldsymbol{\gamma}\cdot\boldsymbol{p} + m^{*}\right\} + \mathrm{g}_{\omega}\omega_{0}\right]\psi = H_{D}\psi}. \\
\Rightarrow \quad & (\psi^{\dagger}H_{D}\psi)_{\boldsymbol{p}} = e(\boldsymbol{p}) = E(\boldsymbol{p}) + \mathrm{g}_{\omega}\omega_{0} \\
\Rightarrow \quad & \frac{\partial}{\partial\zeta}(\psi^{\dagger}H_{D}\psi)_{\boldsymbol{p}} = (\psi^{\dagger}\frac{\partial H_{D}}{\partial\zeta}\psi)_{\boldsymbol{p}} + e(\boldsymbol{p})\frac{\partial}{\partial\zeta}(\psi^{\dagger}\psi)_{\boldsymbol{p}}
\end{aligned}
$$

- Under mean-field approximation σ and ω^{μ} field equations can be further simplified as,

$$
(-\nabla^{2} + m_{\sigma}^{2})\sigma = m_{\sigma}^{2}\langle\sigma\rangle = \mathrm{g}_{\sigma}\langle\overline{\psi}\psi\rangle = \mathrm{g}_{\sigma}\rho_{s} \Rightarrow \boxed{\mathrm{g}_{\sigma}\langle\sigma\rangle = \left(\frac{\mathrm{g}_{\sigma}}{m_{\sigma}}\right)^{2}\langle\overline{\psi}\psi\rangle}.
$$

$$
(-\nabla^{2} + m_{\omega}^{2})\omega^{0}
$$

Simplification of Euler-Lagrange Equations

Dirac equation in Pauli-matrix notation

- We start from the Dirac equation in interaction,

$$
\begin{aligned}
& \left[\gamma^{\mu}\left(i\partial_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}\right) - \left(m - \mathrm{g}_{\sigma}\sigma\right)\right]\psi = 0 \\
\Rightarrow \quad & \left[\gamma^{0}\left(i\partial_{0} - \mathrm{g}_{\omega}\omega_{0}\right) - \boldsymbol{\gamma}\cdot\left(\boldsymbol{p} - \mathrm{g}_{\omega}\boldsymbol{\omega}\right) - m^{*}\right]\psi = 0 \\
\Rightarrow \quad & \boxed{\left[\gamma_{0}\left\{\boldsymbol{\gamma}\cdot\boldsymbol{p} + m^{*}\right\} + \mathrm{g}_{\omega}\omega_{0}\right]\psi = H_{D}\psi}. \\
\Rightarrow \quad & (\psi^{\dagger}H_{D}\psi)_{\boldsymbol{p}} = e(\boldsymbol{p}) = E(\boldsymbol{p}) + \mathrm{g}_{\omega}\omega_{0} \\
\Rightarrow \quad & \frac{\partial}{\partial\zeta}(\psi^{\dagger}H_{D}\psi)_{\boldsymbol{p}} = (\psi^{\dagger}\frac{\partial H_{D}}{\partial\zeta}\psi)_{\boldsymbol{p}} + e(\boldsymbol{p})\frac{\partial}{\partial\zeta}(\psi^{\dagger}\psi)_{\boldsymbol{p}}
\end{aligned}
$$

- Under mean-field approximation σ and ω^{μ} field equations can be further simplified as,

$$
(-\nabla^{2} + m_{\sigma}^{2})\sigma = m_{\sigma}^{2}\langle\sigma\rangle = \mathrm{g}_{\sigma}\langle\overline{\psi}\psi\rangle = \mathrm{g}_{\sigma}\rho_{s} \Rightarrow \boxed{\mathrm{g}_{\sigma}\langle\sigma\rangle = \left(\frac{\mathrm{g}_{\sigma}}{m_{\sigma}}\right)^{2}\langle\overline{\psi}\psi\rangle}.
$$

$$
(-\nabla^{2} + m_{\omega}^{2})\omega^{0} = m_{\omega}^{2}\langle\omega\rangle = \mathrm{g}_{\omega}\langle\psi^{\dagger}\psi\rangle = \mathrm{g}_{\omega}\rho_{b}
$$

Simplification of Euler-Lagrange Equations

Dirac equation in Pauli-matrix notation

- We start from the Dirac equation in interaction,

$$
\begin{aligned}
& \left[\gamma^{\mu}\left(i\partial_{\mu} - \mathrm{g}_{\omega}\omega_{\mu}\right) - \left(m - \mathrm{g}_{\sigma}\sigma\right)\right]\psi = 0 \\
\Rightarrow \quad & \left[\gamma^{0}\left(i\partial_{0} - \mathrm{g}_{\omega}\omega_{0}\right) - \boldsymbol{\gamma}\cdot\left(\boldsymbol{p} - \mathrm{g}_{\omega}\boldsymbol{\omega}\right) - m^{*}\right]\psi = 0 \\
\Rightarrow \quad & \boxed{\left[\gamma_{0}\left\{\boldsymbol{\gamma}\cdot\boldsymbol{p} + m^{*}\right\} + \mathrm{g}_{\omega}\omega_{0}\right]\psi = H_{D}\psi}. \\
\Rightarrow \quad & (\psi^{\dagger}H_{D}\psi)_{\boldsymbol{p}} = e(\boldsymbol{p}) = E(\boldsymbol{p}) + \mathrm{g}_{\omega}\omega_{0} \\
\Rightarrow \quad & \frac{\partial}{\partial\zeta}(\psi^{\dagger}H_{D}\psi)_{\boldsymbol{p}} = (\psi^{\dagger}\frac{\partial H_{D}}{\partial\zeta}\psi)_{\boldsymbol{p}} + e(\boldsymbol{p})\frac{\partial}{\partial\zeta}(\psi^{\dagger}\psi)_{\boldsymbol{p}}
\end{aligned}
$$

- Under mean-field approximation σ and ω^{μ} field equations can be further simplified as,

$$
(-\nabla^{2} + m_{\sigma}^{2})\sigma = m_{\sigma}^{2}\langle\sigma\rangle = \mathrm{g}_{\sigma}\langle\overline{\psi}\psi\rangle = \mathrm{g}_{\sigma}\rho_{s} \Rightarrow \boxed{\mathrm{g}_{\sigma}\langle\sigma\rangle = \left(\frac{\mathrm{g}_{\sigma}}{m_{\sigma}}\right)^{2}\langle\overline{\psi}\psi\rangle}.
$$

$$
(-\nabla^{2} + m_{\omega}^{2})\omega^{0} = m_{\omega}^{2}\langle\omega\rangle = \mathrm{g}_{\omega}\langle\psi^{\dagger}\psi\rangle = \mathrm{g}_{\omega}\rho_{b} \Rightarrow \boxed{\mathrm{g}_{\omega}\langle\omega\rangle = \left(\frac{\mathrm{g}_{\omega}}{m_{\omega}}\right)^{2}\langle\psi^{\dagger}\psi\rangle}.
$$

Mean Field Equations

Self consistency!

Mean Field Equations

Self consistency!

- First we define expectation value of an operator Γ in many body nucleon state in ground state as

$$\langle\overline{\psi}\Gamma\psi\rangle = \sum_{\kappa}\int\frac{d\boldsymbol{p}}{(2\pi)^3}(\overline{\psi}\Gamma\psi)_{\boldsymbol{p}_\kappa}\Theta[\mu - \mathrm{e}(\boldsymbol{p})].$$

Mean Field Equations

Self consistency!

- First we define expectation value of an operator Γ in many body nucleon state in ground state as

$$\langle\overline{\psi}\Gamma\psi\rangle = \sum_{\kappa}\int\frac{d\boldsymbol{p}}{(2\pi)^3}(\overline{\psi}\Gamma\psi)_{\boldsymbol{p}_\kappa}\Theta[\mu - \mathrm{e}(\boldsymbol{p})].$$

$$\Rightarrow \quad \langle\overline{\psi}\psi\rangle = \frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\frac{m - \mathrm{g}_\sigma\sigma}{\sqrt{p^2 + (m - \mathrm{g}_\sigma\sigma)^2}}, \text{ as, } (\overline{\psi}\psi)_{\boldsymbol{p}} = \frac{\partial E(p)}{\partial m}.$$

Mean Field Equations

Self consistency!

- First we define expectation value of an operator Γ in many body nucleon state in ground state as

$$\langle\overline{\psi}\Gamma\psi\rangle = \sum_{\kappa}\int\frac{d\boldsymbol{p}}{(2\pi)^3}(\overline{\psi}\Gamma\psi)_{\boldsymbol{p}_\kappa}\Theta[\mu - \mathrm{e}(\boldsymbol{p})].$$

$$\Rightarrow \quad \langle\overline{\psi}\psi\rangle = \frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\frac{m - \mathrm{g}_\sigma\sigma}{\sqrt{p^2 + (m - \mathrm{g}_\sigma\sigma)^2}}, \text{ as, } (\overline{\psi}\psi)_{\boldsymbol{p}} = \frac{\partial E(p)}{\partial m}.$$

- So the field equations can be written as,

$$\mathrm{g}_\sigma\langle\sigma\rangle = \left(\frac{\mathrm{g}_\sigma}{m_\sigma}\right)^2\langle\overline{\psi}\psi\rangle = \left(\frac{\mathrm{g}_\sigma}{m_\sigma}\right)^2\frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\frac{m - \mathrm{g}_\sigma\sigma}{\sqrt{p^2 + (m - \mathrm{g}_\sigma\sigma)^2}},$$

$$\mathrm{g}_\omega\langle\omega\rangle = \left(\frac{\mathrm{g}_\omega}{m_\omega}\right)^2\langle\psi^\dagger\psi\rangle = \left(\frac{\mathrm{g}_\omega}{m_\omega}\right)^2\rho_b.$$

Mean Field Equations

Self consistency!

- First we define expectation value of an operator Γ in many body nucleon state in ground state as

$$\langle\overline{\psi}\Gamma\psi\rangle = \sum_{\kappa}\int\frac{d\boldsymbol{p}}{(2\pi)^3}(\overline{\psi}\Gamma\psi)_{\boldsymbol{p}_\kappa}\Theta[\mu - \mathrm{e}(\boldsymbol{p})].$$

$$\Rightarrow \quad \langle\overline{\psi}\psi\rangle = \frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\frac{m - \mathrm{g}_\sigma\sigma}{\sqrt{p^2 + (m - \mathrm{g}_\sigma\sigma)^2}}, \text{ as, } (\overline{\psi}\psi)_{\boldsymbol{p}} = \frac{\partial E(p)}{\partial m}.$$

- So the field equations can be written as,

$$\mathrm{g}_\sigma\langle\sigma\rangle = \left(\frac{\mathrm{g}_\sigma}{m_\sigma}\right)^2\langle\overline{\psi}\psi\rangle = \left(\frac{\mathrm{g}_\sigma}{m_\sigma}\right)^2\frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\frac{m - \mathrm{g}_\sigma\sigma}{\sqrt{p^2 + (m - \mathrm{g}_\sigma\sigma)^2}},$$

$$\mathrm{g}_\omega\langle\omega\rangle = \left(\frac{\mathrm{g}_\omega}{m_\omega}\right)^2\langle\psi^\dagger\psi\rangle = \left(\frac{\mathrm{g}_\omega}{m_\omega}\right)^2\rho_b.$$

- $m_\sigma, m_\omega, \mathrm{g}_\sigma, \mathrm{g}_\omega, m$ are provided.

Mean Field Equations

Self consistency!

- First we define expectation value of an operator Γ in many body nucleon state in ground state as

$$\langle\overline{\psi}\Gamma\psi\rangle = \sum_{\kappa}\int\frac{d\boldsymbol{p}}{(2\pi)^3}(\overline{\psi}\Gamma\psi)_{\boldsymbol{p}_\kappa}\Theta[\mu - \mathrm{e}(\boldsymbol{p})].$$

$$\Rightarrow \quad \langle\overline{\psi}\psi\rangle = \frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\frac{m - \mathrm{g}_\sigma\sigma}{\sqrt{p^2 + (m - \mathrm{g}_\sigma\sigma)^2}}, \text{ as, } (\overline{\psi}\psi)_{\boldsymbol{p}} = \frac{\partial E(p)}{\partial m}.$$

- So the field equations can be written as,

$$\mathrm{g}_\sigma\langle\sigma\rangle = \left(\frac{\mathrm{g}_\sigma}{m_\sigma}\right)^2\langle\overline{\psi}\psi\rangle = \left(\frac{\mathrm{g}_\sigma}{m_\sigma}\right)^2\frac{2}{\pi^2}\int_0^{k_f} p^2\,dp\frac{m - \mathrm{g}_\sigma\sigma}{\sqrt{p^2 + (m - \mathrm{g}_\sigma\sigma)^2}},$$

$$\mathrm{g}_\omega\langle\omega\rangle = \left(\frac{\mathrm{g}_\omega}{m_\omega}\right)^2\langle\psi^\dagger\psi\rangle = \left(\frac{\mathrm{g}_\omega}{m_\omega}\right)^2\rho_b.$$

- $m_\sigma, m_\omega, \mathrm{g}_\sigma, \mathrm{g}_\omega, m$ are provided. For a given ρ_b these are solved self consistently.

References:

- Compact Stars, Springer, **Norman K. Glendenning**.
- Annals of Physics 83, 491-529 (1974), **J. D. Walecka**.
- Quantum Field Theory, Cambridge University Press, **Lewis H. Ryder**.